

CT and MR Enterography in the Evaluation of Celiac Disease

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Celiac disease is a common inflammatory disease of the small bowel that induces mucosal intestinal lesions. The disease is mediated by an immune response and triggered by the ingestion of gluten in genetically predisposed individuals. Gluten contains gliadin, a component found mostly in wheat, barley, and rye. This process leads to gastrointestinal malabsorption with symptoms such as diarrhea, constipation, abdominal pain, and distention. It has a prevalence of 1%–2% in the general adult population, who present with symptoms at any age, but is more frequently found in adult women in the 3rd or 4th decade of life. Recognition of the disease has increased, but it remains a challenge to diagnose. CT and MR enterography are noninvasive studies used for evaluation of small bowel neoplasms and inflammatory small bowel pathologic conditions such as celiac disease. The authors review the spectrum of intestinal and extraintestinal findings of celiac disease at CT and MR enterography, as well as its complications, and the importance of recognizing certain imaging features that help in the diagnosis of celiac disease. More common and specific findings of celiac disease such as inversion of the jejunoileal fold pattern and mesenteric lymphadenopathy are reviewed. More uncommon entities that are more frequently associated with refractory or untreated celiac disease, such as ulcerative jejunoileitis, cavitory mesenteric lymph node syndrome, and malignancies including small bowel adenocarcinoma and lymphoma, are described.

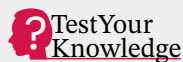
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Introduction

Celiac disease, also known as celiac sprue or gluten-sensitive enteropathy, is a common inflammatory disease of the small bowel that is characterized by the loss of villi in the proximal small bowel. It is mediated by an immune response triggered by the ingestion of gluten in genetically predisposed individuals. Gluten refers to a family of proteins from wheat, barley, rye, and their cross-bred grains and also oats in rare cases (1,2). This process leads to gastrointestinal malabsorption

with symptoms of diarrhea, constipation, and abdominal pain and distention (3). The diagnosis of celiac disease must always be confirmed by biopsy. Typical histologic features are increased intraepithelial lymphocytes (IELs), crypt hyperplasia, and villous atrophy.

At CT and MR enterography, important findings such as an abnormal fold pattern and mesenteric lymphadenopathy are characteristic findings. Other imaging features include duodenitis, small bowel dilatation, jejunoileal fold pattern reversal,



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Abbreviations: FDG = fluorodeoxyglucose, H-E = hematoxylin-eosin, HLA = human leukocyte antigen, IEL = intraepithelial lymphocyte, IL = interleukin, IgA = immunoglobulin A, TG2 = transglutaminase 2

TEACHING POINTS

- Inversion of the jejunoileal fold pattern is the most specific sign of celiac disease and is described in the literature as a finding approaching 100% sensitivity and specificity (positive and negative predictive values).
- Intussusception or multiple synchronous intussusceptions without lead point masses or obstruction should alert the radiologist to the possibility of undiagnosed celiac disease in asymptomatic adults.
- Splenic atrophy is often associated with the presence of cavitated lymph nodes and, when seen together, can increase diagnostic specificity for celiac disease.
- Ulcerative jejunoileitis may lead to severe stricture or hemorrhage, perforation, and obstruction.
- At imaging, cavitory mesenteric lymph node syndrome (CMLNS) can manifest as numerous enlarged mesenteric lymph nodes that vary in size (2–8 cm in diameter) and are centrally low in attenuation, signifying their centrally lipid or hyaline-type composition. Some lymph nodes may demonstrate a fat-fluid level, which is highly specific for CMLNS in celiac disease.

and transient intussusception. Untreated celiac disease leads to chronic inflammation of the small intestine, which increases the risk of small bowel neoplasms, strictures, and progression to refractory celiac disease. This article reviews the diagnosis of celiac disease, with a particular focus on CT and MR enterographic findings of celiac disease and its complications.

Epidemiologic Risk Factors

Celiac disease has a prevalence of 1%–2% in the general adult population (4–6). The prevalence of celiac disease in pediatric patients, according to studies conducted in the United States, ranges from 0.2% to 1% (7). Approximately 3 million people are affected in Europe and at least another 3 million in the United States (1). Celiac disease is also relatively common in Asia, with a prevalence approaching 1.5% (8). The disease can occur in patients at any age, but it is more frequently found in adult women in the 3rd or 4th decade of life, with a female-to-male ratio from 2:1 to 3:1 (9).

The global distribution of the disease parallels the distribution of human leukocyte antigen (HLA) genotypes that predispose an individual to celiac disease, provided that the population is also exposed to gluten (10). Population-based studies suggest that recognized cases of celiac disease represent a small part of the total number of patients with this disease (10). Experts describe the prevalence of this disease as akin to an iceberg. The area of the iceberg above water refers to clinically diagnosed celiac disease, while the area of the iceberg below the water represents the total number of undiagnosed cases in a given population at a given time (Fig 1) (11).

Among the groups at high risk are first- and second-degree relatives, with a prevalence of 8.9% for siblings, 7.9% for children, and 3% for parents of the celiac patient (10). Celiac disease is associated with several autoimmune diseases, especially

type 1 diabetes and Hashimoto thyroiditis. It is less frequently associated with autoimmune hepatitis infection, primary biliary cholangitis, systemic lupus erythematosus, Sjögren syndrome, systemic sclerosis, and inflammatory bowel disease. Some authors (12) have suggested that gluten may also trigger autoimmune disorders. It is possible to find immunoglobulin A (IgA) deficiency in patients with genetic alterations and chromosomopathies such as Down syndrome, Turner syndrome, and Williams syndrome (12).

There is also a spectrum of gluten sensitivity entities, characterized by intestinal symptoms related to the ingestion of gluten in people who are not affected by celiac disease, called *non-celiac gluten sensitivity* (NCGS) or wheat allergy (13). Unlike celiac disease, NCGS is not an autoimmune process that leads to intestinal epithelial cell damage. Celiac disease is a common but underdiagnosed disease, so a high index of suspicion is necessary for diagnosis.

Pathogenesis

Celiac disease is an autoimmune inflammatory condition that is triggered in genetically predisposed individuals by the ingestion of the gluten proteins from wheat, barley, rye, and their cross-bred grains and, in rare cases, oats (14). There is a close association with the *HLA-DR3/DQ2* and *HLA-DR4/DQ8* gene locus in patients with celiac disease compared with that in the general population (15). Ingested gluten is partially digested by host (digestive) and microbial (small intestinal microbiota) enzymes and then transported through transcellular and paracellular pathways to the lamina propria. Activated transglutaminase 2 (TG2), an enzyme secreted during inflammatory conditions, deamidates glutamine residues on gluten peptides, which improves the efficiency of peptide binding to the *HLA-DQ2* or *HLA-DQ8* groove of antigen-presenting cells. TG2-gluten complexes activate B cells that produce autoantibodies, including antibodies against TG2 that are used as a sensitive and specific serologic diagnostic tool.

T-helper cells secrete proinflammatory mediators such as interferon γ , interleukin (IL)-12, and tumor necrosis factor. B cells can also internalize TG2-gluten complexes and release deamidated gluten peptides that bind to *HLA-DQ2* or *HLA-DQ8*, further activating gluten-specific T cells. Infectious or inflammatory signals upregulate stress markers and histocompatibility antigen HLA class I in intestinal epithelial cells. IELs interact with intestinal epithelial cells through natural killer cell receptors. Proinflammatory cytokines such as IL-15 drive IEL cytotoxicity, leading to mucosal destruction and intestinal atrophy (Fig 2) (12).

Clinical Manifestations

Diagnosis of celiac disease can be difficult, as many symptoms are nonspecific. Currently the diagnosis is suspected in patients with diarrhea and other complementary signs or symptoms such as abdominal pain, bloating, vomiting, anemia, or osteoporosis (16). According to the Oslo classification of 2011, there are multiple definitions related to celiac disease that divide celiac disease into classic, nonclassic, subclinical, refractory, latent, and potential manifestations. The classic presentation of a patient with celiac disease is related to

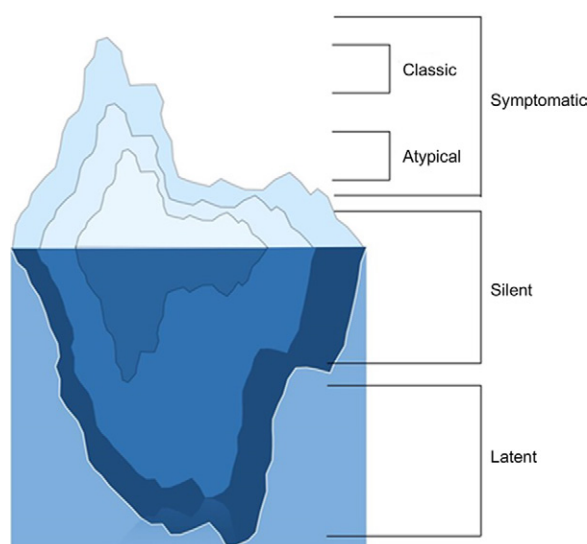


Figure 1. Iceberg model depicts the different forms of manifestation of celiac disease: symptomatic, with typical or atypical symptoms; silent, an asymptomatic form with the presence of specific antibodies; and latent, defined by the presence of typical disease genes but with no evidence of intestinal damage; patients may or may not have specific antibodies or symptoms.

gastrointestinal malabsorption that manifests with diarrhea, steatorrhea, or weight loss. Patients with nonclassic celiac disease present without signs and symptoms of malabsorption, and patients can present with just abdominal pain and constipation (17,18). Nevertheless, a more practical approach is to describe the presentation of patients with celiac disease as typical (those with gastrointestinal symptoms) or atypical (those with extraintestinal symptoms) (14).

Patients with gastrointestinal manifestations are more frequently in the pediatric population and may present with classic signs related to malabsorption (19). Extraintestinal manifestations are frequently seen in children and adults and include mucocutaneous abnormalities, metabolic bone disorders, hematologic abnormalities, and neurologic manifestations (Table 1) (20).

Associated Conditions

Celiac disease can be associated with immune, idiopathic, and oncologic pathologic conditions (Table 2). Intestinal dysfunction associated with undiagnosed celiac disease increases intestinal permeability to certain triggers, which may promote the onset of other autoimmune disorders. Risk increases with age and the length of gluten exposure (21,22). Associated autoimmune diseases include type 1 diabetes mellitus, Hashimoto thyroiditis, Graves disease, primary biliary cholangitis, vitiligo, Addison disease, IgA deficiency, and rheumatologic diseases (5). Other associated idiopathic entities include dilated cardiomyopathy, cerebellar ataxia, multiple myoclonic seizures, multiple sclerosis, and cerebral atrophy (6,21).

Diagnostic Approach

Patients are screened for celiac disease when clinical suspicion arises. In addition, the American College of Gastro-

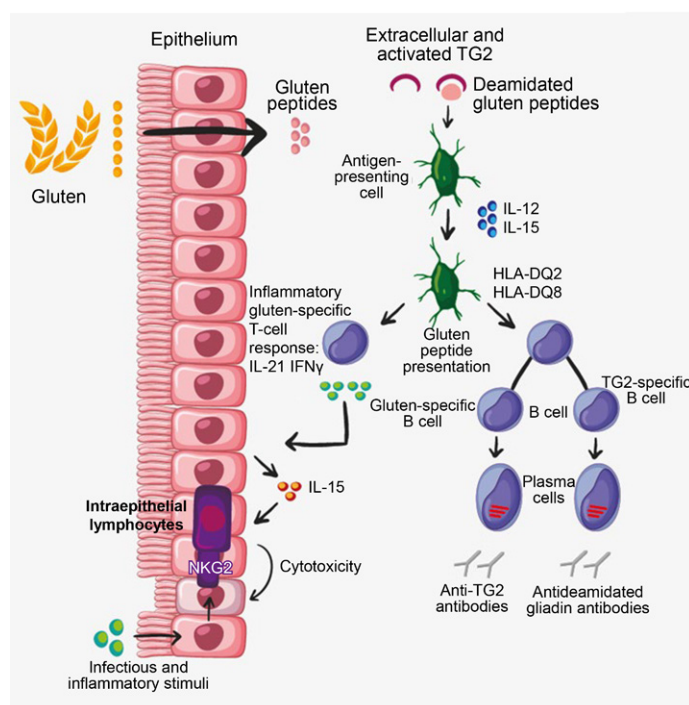


Figure 2. Diagram shows the pathogenesis of celiac disease. *IFN γ* = interferon γ , *NKG2* = natural killer cell receptor.

enterology (ACG) clinical guidelines on celiac disease state that testing should be considered when a first-degree relative has biopsy-confirmed disease even in the absence of symptoms (23).

The diagnosis of celiac disease is established with the combination of pathologic small bowel changes in addition to serologic test results. The first-line serologic test in adults is the IgA tissue transglutaminase antibody test, which carries a 95% sensitivity and specificity (24). Patients with an underlying IgA deficiency may produce a false-negative serum result (21). In such patients, an immunoglobulin-G antideamidated gliadin peptide test can be performed (4). Antigliadin antibody tests are no longer recommended because of the inferior results compared with those of other tests. Serologic screening is recommended by the ACG in asymptomatic first-degree relatives of patients with celiac disease (4).

Adults with positive serologic results often have a diagnosis of celiac disease that is confirmed using tissue sampling performed during endoscopic small bowel biopsy. However, in the pediatric population, the European Society for Pediatric Gastroenterology, Hepatology and Nutrition guidelines propose allowing a nonbiopsy diagnosis of celiac disease in patients with a positive transglutaminase antibody greater than 10 times the upper limit of a normal and positive endomysial antibody test result and a positive *HLA-DQ2* or *HLA-DQ8* test result (25).

In adult and pediatric patients, a negative serologic result does not exclude celiac disease, as this result is seen with a reduction or elimination of gluten from the diet, selective IgA deficiency, use of corticosteroids, or immunomodulating medication. If there is high clinical suspicion, biopsy of the small bowel is often performed (26).

Table 1: Extraintestinal Manifestations of Celiac Disease

Type	Manifestations
Mucocutaneous	Dermatitis herpetiformis is the most frequent condition; it is characterized by a pruritic, papular, and vesicular rash with deposits of IgA at biopsy (51) Other less common mucocutaneous manifestations are eczema, psoriasis, or atrophic glossitis (10)
Metabolic bone disorders	Chronic inflammation of the small bowel leads to malabsorption of calcium and vitamin D, resulting in metabolic bone disorders and increased risk of fractures (52) Osteomalacia (53)
Hematologic	Small bowel inflammation and ulceration may lead to small bowel hemorrhage, increasing the risk of iron deficiency anemia (54)
Neurologic	The pathogenesis of neurologic symptoms is unclear, but different mechanisms have been suggested such as nutritional, immunologic, toxic, and metabolic disturbances (55) Headache is the most common manifestation; anxiety, depression, ataxia, epilepsy, and peripheral neuropathy; neuropathy is related to vitamin B deficiency or associated lymphoma (56)
Other	Hypertransaminasemia, infertility, late menarche, amenorrhea, and early menopause (57,58)

Table 2: Conditions Associated with Celiac Disease

Type	Associated Disorders
Autoimmune	Type 1 diabetes mellitus, Hashimoto thyroiditis, Graves disease, primary biliary cholangitis, vitiligo, Addison disease, IgA deficiency, and rheumatologic disease
Idiopathic	Dilated cardiomyopathy, cerebellar ataxia, multiple myoclonic seizures, multiple sclerosis, and cerebral atrophy (6,21)
Oncologic	Lymphoma, esophageal adenocarcinoma, and small bowel carcinoma (59); in one of the largest population studies, Askling et al (60) concluded that the cancer occurrence is temporarily elevated, not after 10 or more years from initial hospitalization
Mortality	Compared with the general population, people with celiac disease have an increased risk of mortality, mainly owing to respiratory disease, lymphoproliferative diseases, and non-Hodgkin lymphoma, with a more prominent risk when the disease is diagnosed after age 40 years (61,62)

Table 3: Advantages and Limitations of CT and MR Enterography in Evaluation of Celiac Disease

Factor	CT Enterography	MR Enterography
Advantages	Better spatial resolution Global overview of abdomen Better for recognizing ulcerative jejunoileitis and cavitary mesenteric lymph nodes	Better tissue resolution No exposure to ionizing radiation; recommended for pediatric patients or those requiring continuous follow-up imaging Ability to perform the examination in patients with impaired renal function with MRI contrast agents Better for recognizing inversion of the jejunoileal fold pattern and intestinal wall thickening
Limitations	Exposure to ionizing radiation Necessity of using intravenous iodine contrast agent and contraindicated in patients with impaired renal function	Longer examination and interpretation times Susceptible to motion and breathing artifacts More expensive

Imaging Features of Celiac Disease

Celiac disease has a wide spectrum of intestinal and extraintestinal imaging findings that are related to the stage or severity of the disease. Table 3 summarizes some advantages and limitations of CT and MR enterography in evaluation of celiac disease. Although many of the intestinal findings are nonspecific, the presence of several of these findings should lead to a more careful examination for other intestinal and extraintestinal findings of celiac disease.

Intestinal Findings

Abnormalities in Intestinal Fold Pattern.—The imaging hallmark of celiac disease is an alteration of the small bowel fold pattern (Fig 3). Chronic inflammation initiated by gluten leads to villous atrophy and appears as decreased small bowel folds on CT and MR images (Fig 4). The extent of disease is extremely variable and ranges from segmental involvement to diffuse involvement of the small bowel, but most commonly it affects the proximal jejunum (3). As the inflammatory

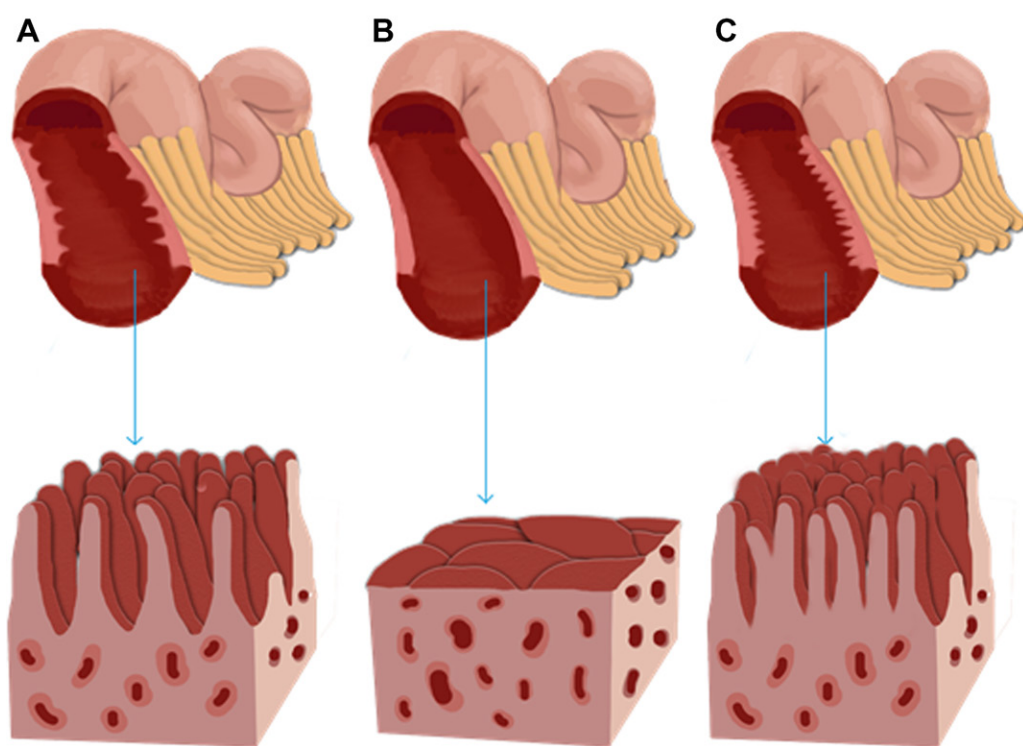


Figure 3. Illustrations show the normal villous appearance (A), villous atrophy caused by celiac disease (B), and an increase in the number of folds and villi in the ileum (C).

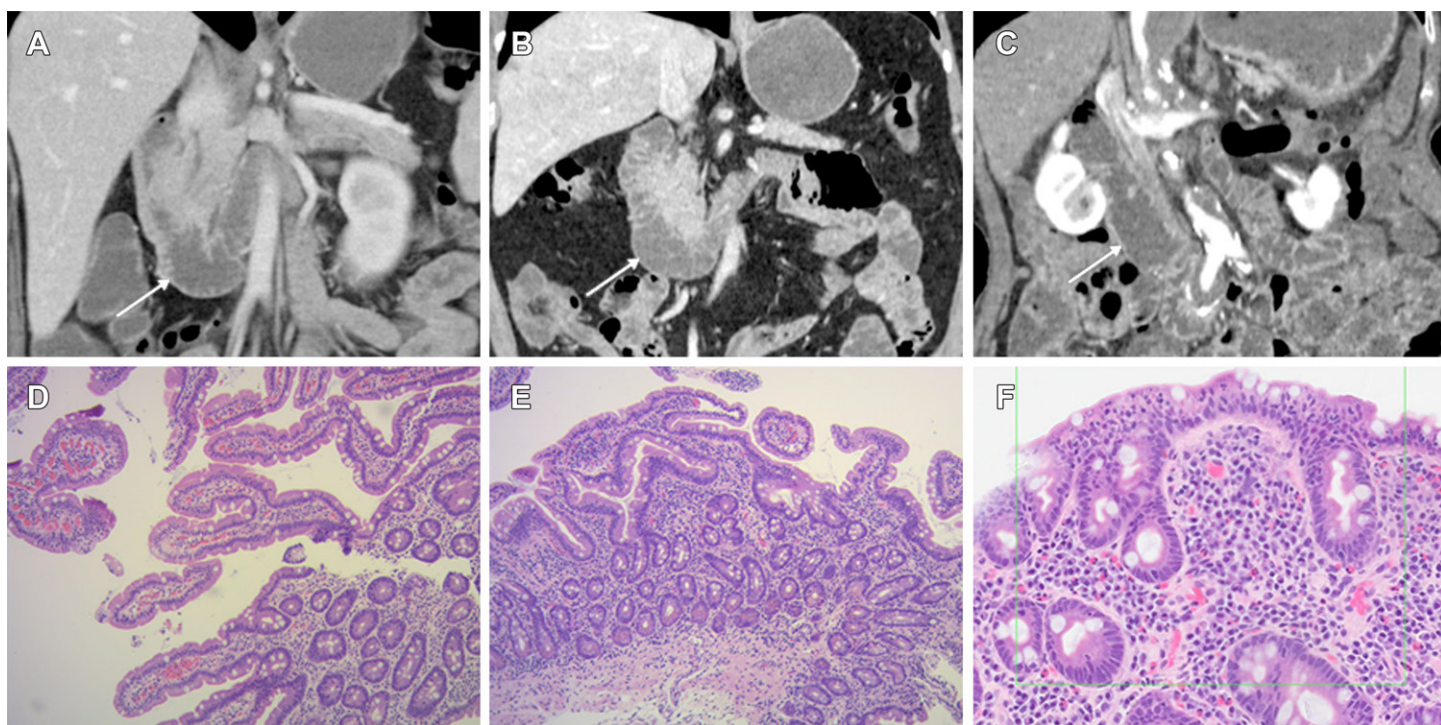


Figure 4. Celiac disease at CT enterography with histologic correlation in three patients. (A–C) Coronal intravenous contrast-enhanced CT enterographic images show a normal fold pattern of the duodenum (arrow in A), a reduced number of duodenal folds (arrow in B), and total loss of duodenal folds (arrow in C). (D) Photomicrograph at histologic evaluation shows normal villous development and a normal IEL count. (Hematoxylin-eosin [H-E] stain; original magnification, $\times 10$.) (E) Photomicrograph shows partial patchy villous atrophy and increased IELs ($>30\%$). (H-E stain; original magnification, $\times 10$.) (F) Photomicrograph shows total villous atrophy and intraepithelial lymphocytosis at histologic evaluation. (H-E stain; original magnification, $\times 20$.)

process progresses, destruction and flattening of the mucosa and thickening of the wall and folds are generated (1), which can present with four imaging patterns: normal fold

pattern, folds with square ends (margins of the small bowel folds are squared instead of rounded), inversion of the fold pattern between the jejunum and ileum (called jejunoization



Figure 5. Celiac disease in a 59-year-old woman. Coronal intravenous contrast-enhanced CT enterographic images show inversion of the small bowel fold pattern manifested by a decrease in the number of folds in the jejunum (arrow in **A**) due to villous atrophy and an increase in the number of folds in the ileum (arrowheads in **B**) representing jejunization of the ileum as a compensation for malabsorption.

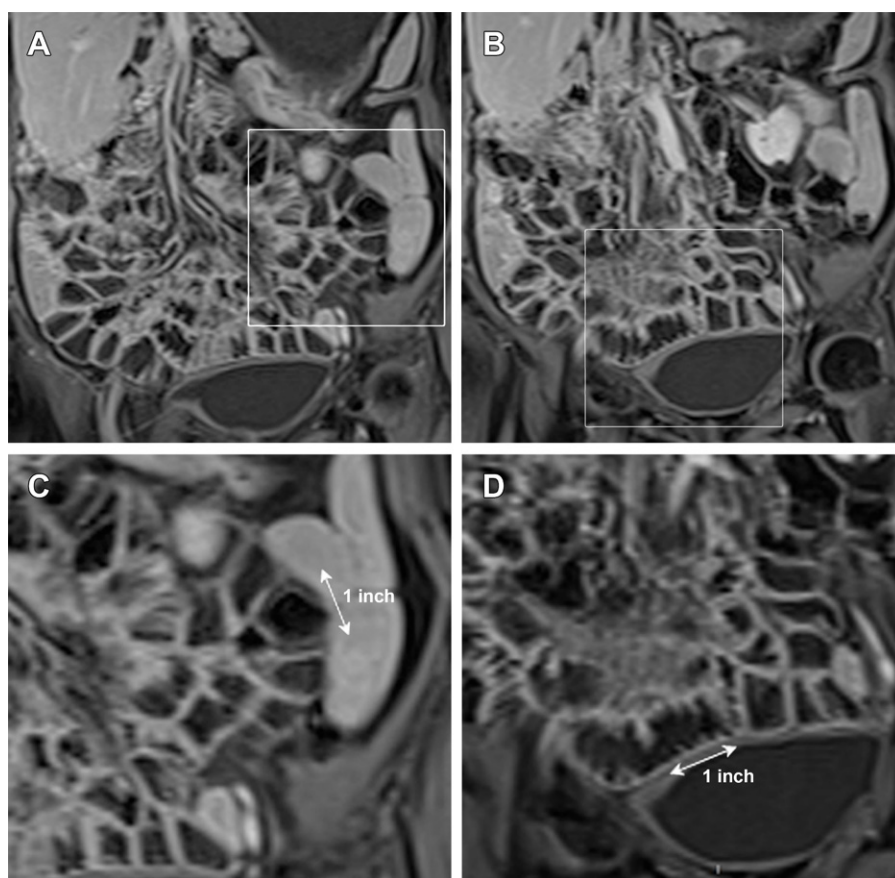


Figure 6. MR enterographic evaluation of fold patterns. (**A, B**) Coronal intravenous contrast-enhanced fat-suppressed T1-weighted MR enterographic images show a decreased number of folds in the jejunum (**A**) and an increased number of folds in the ileum (**B**). (**C, D**) Magnification images of the jejunum (**C**) and ileum (**D**) better depict the inversion of the fold patterns. Double-headed arrow denotes the number of folds per inch.

of the ileum, with decreased jejunal folds and increased ileal folds) (Fig 5), and total absence of the folds (“moulage” sign with total villous atrophy) (14).

Jejunal folds are considered to be decreased in density if there are three or fewer folds per 1-inch length of bowel seen on CT and MR enterographic images (Fig 6). In severe celiac disease, there may be a complete loss of jejunal folds (14). With chronic inflammation, there may be a compensatory

increase in ileal fold density, defined as more than five folds per 1 inch, in an attempt to increase the small bowel surface area (3). The increased number of folds represents an adaptation of the ileum to the reduced absorptive capacity of the jejunum (14).

Inversion of the jejunoileal fold pattern is the most specific sign of celiac disease and is described in the literature as a finding approaching 100% sensitivity and specificity

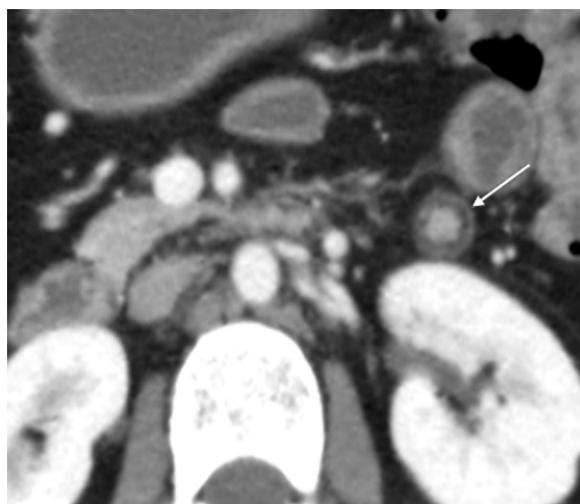


Figure 7. Celiac disease in a 40-year-old woman who complied with a gluten-free diet and presented with findings of anemia and no response to iron therapy. Axial intravenous contrast-enhanced CT enterographic image shows mural thickening of jejunal loops and fat deposition (arrow) of the submucosa due to chronic inflammation.

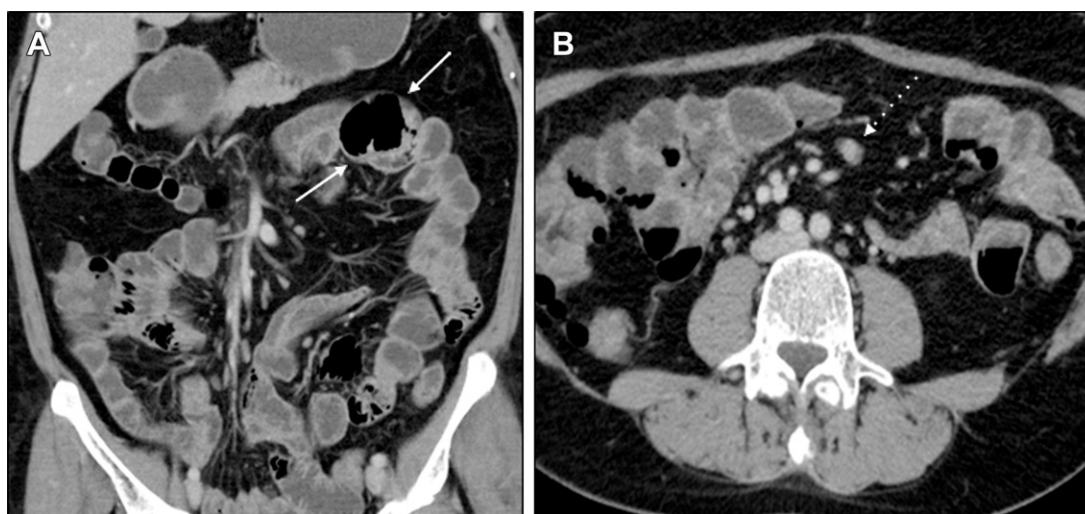


Figure 8. Celiac disease in a 59-year-old asymptomatic woman with a history of hospitalization the previous year for intestinal obstruction without surgical intervention. Coronal (A) and axial (B) intravenous contrast-enhanced CT enterographic images show focal dilatation of 32 mm in a jejunal loop (arrows in A), with no wall thickening and no evidence of a stricture. There are multiple homogeneous and slightly enlarged mesenteric nodes (arrow in B) caused by lymphocyte stimulation.

(positive and negative predictive values) (27). The duration of diarrhea and malabsorption tends to be longer in patients with jejunoileal fold pattern reversal than that in patients with simple jejunization of the ileum. After treatment with a gluten-free diet, the recovery of a normal intestinal mucosal fold pattern is determined by the degree of villous atrophy and severity of disease at the time of treatment initiation. Patients with a more severe presentation and/or total villous atrophy have lower recovery rates, and those with increased recovery rates have a lower risk of complications and lower mortality rates. A median time of approximately 3.8 years is suggested to confirm mucosal recovery at biopsy in these patients (28).

Intestinal Wall Thickening.—Intestinal wall and fold thickening are often the first radiologic sign of celiac disease, preceding abnormalities in the fold pattern, and are likely the result of edema and inflammation from submucosal lymphocytic infiltration. When intestinal wall thickening is present with other more specific findings of celiac disease, the likelihood of celiac disease is increased.

Intestinal wall thickening is defined as a thickness of more than 3 mm and requires optimal luminal distention (Fig 7).

A poorly distended small bowel can lead to a false-positive result, as it may mimic bowel wall thickening or mucosal hyperenhancement. In addition to the inflammation caused by celiac disease itself, the destruction of villi in the jejunum predisposes the more proximal small bowel to become inflamed by gastric, biliary, and pancreatic secretions (14). Generally, the thickening of the intestinal wall is diffuse and is not associated with a reduction in intestinal caliber (3).

Intestinal Dilatation.—Dilatation of the small bowel is a non-specific feature of celiac disease (29). It has a predilection for the jejunum and is thought to occur secondary to intestinal hypomotility. Small bowel dilatation at CT or MR enterography is defined by a luminal diameter greater than 3 cm, with preserved wall thickness (Fig 8) (3). In some cases, intestinal dilatation may cause effacement of the folds. In patients with celiac disease, intestinal dilatation does not often cause effacement of the intestinal folds, and jejunization of the ileum can still be recognized in dilated ileal loops (30).

Intestinal dilatation may also simulate intestinal obstruction, paralytic ileus, or other rare entities such as intestinal scleroderma. In these cases, the patient's clinical history is crucial to help determine the most likely process. In the absence

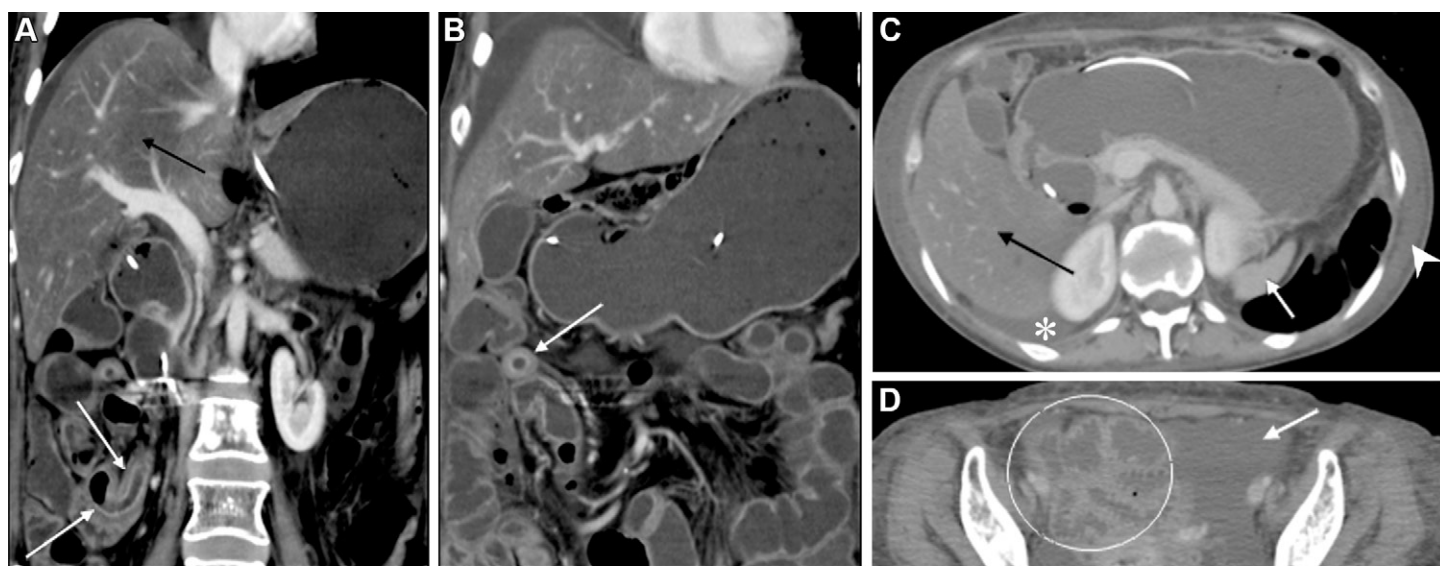


Figure 9. Recently diagnosed celiac disease in a 45-year-old woman with chronic anemia, weight loss, hematochezia, and proctorrhagia. **(A, B)** Coronal intravenous contrast-enhanced CT enterographic images show a long stricture involving the terminal ileum and ileocecal valve, associated with bilaminar stratified hyperenhancement and hyperemia with prominence of the vasa recta (white arrows). Note the low attenuation of the liver due to fatty deposition or edema (black arrow in **A**). **(C)** Axial CT image shows right pleural effusion (*), subcutaneous edema (arrowhead), hyposplenism (white arrow), and low attenuation of the liver (black arrow). **(D)** Axial CT image shows ascites (arrows) and inversion of the fold pattern (circle). The patient was hospitalized, and parenteral feeding was performed until clinical improvement was observed.

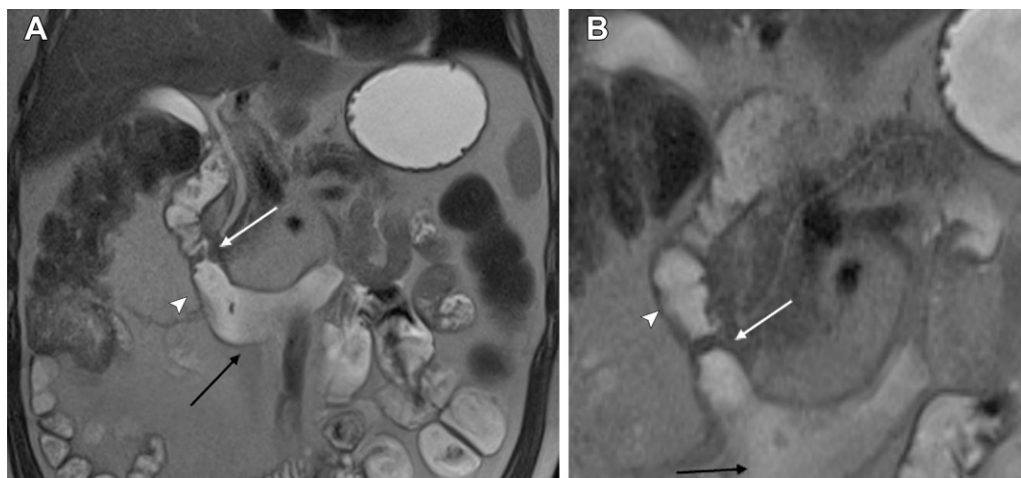


Figure 10. Recently diagnosed celiac disease in a 37-year-old woman. Coronal T2-weighted MR enterographic images show loss of duodenal folds (arrowhead) compatible with villous atrophy, as well as a duodenal stricture (white arrow) associated with mild ectasia of the third portion of the duodenum (black arrow).

of imaging findings or a clinical history compatible with obstructive disease, drug-induced hypotonia, or acute intestinal infection, celiac disease should be considered in patients with intestinal dilatation and no underlying cause (29).

Strictures.—Isolated intestinal strictures are an uncommon and nonspecific sign in celiac disease (3). They manifest as segmental luminal narrowing with or without upstream bowel dilatation (Figs 9A, 9B, 10) (14). If there is active inflammation in the stenotic region, segmental hyperenhancement of the bowel wall occurs. Enhancement can be stratified (bilaminar when it enhances only the mucosa and trilaminar when it enhances the mucosa and serosa, accompanied by edema of the submucosa) or homogeneous (28). These types of findings could be confused with Crohn disease, but the strictures

in Crohn disease tend to be multiple and segmental and are uncommon in celiac disease.

Intussusception.—Intussusception is due to a dilated and atonic or hypotonic small bowel with abnormal peristalsis, resulting in invagination of one segment of small bowel into another (31). It is a complication that may be incidental and asymptomatic but may also be a cause of severe abdominal pain. Intussusception or multiple synchronous intussusceptions without lead point masses or obstruction should alert the radiologist to the possibility of undiagnosed celiac disease in asymptomatic adults.

At CT and MRI, intussusception is characterized by a double halo with alternating layers of the intestinal wall. The innermost layer represents the hyperenhancing mucosa, followed

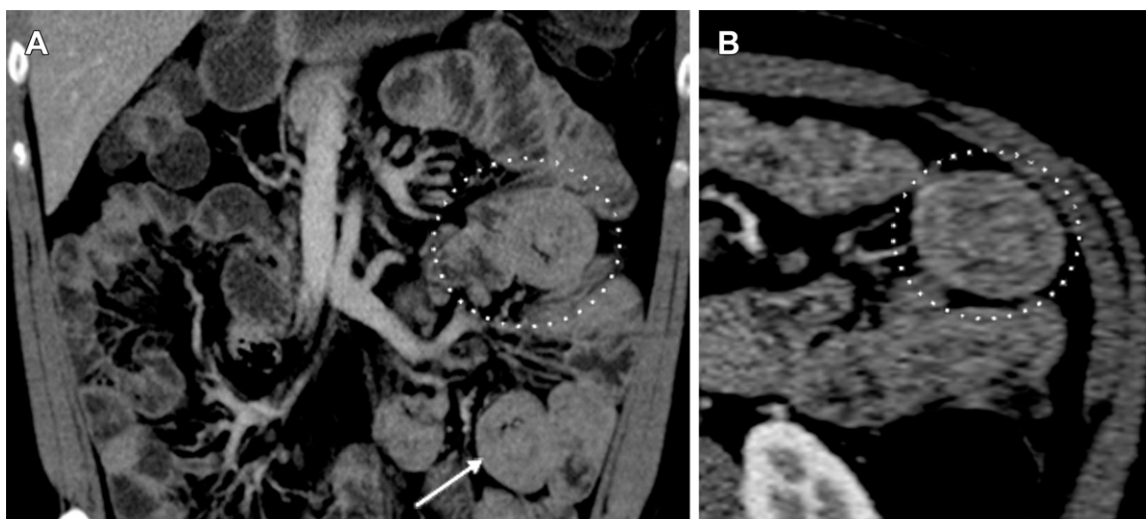


Figure 11. Recently diagnosed celiac disease in a 37-year-old man with diarrhea and fever. Coronal intravenous contrast-enhanced CT image (**A**) shows two short segments of small bowel intussusception (arrow in **A**) in the left hemiabdomen. The patient was discharged from the hospital with no surgical procedure due to favorable evolution. Axial magnified intravenous contrast-enhanced enterographic CT image (**B**) demonstrates the left superior intussusception seen in **A** (circle).

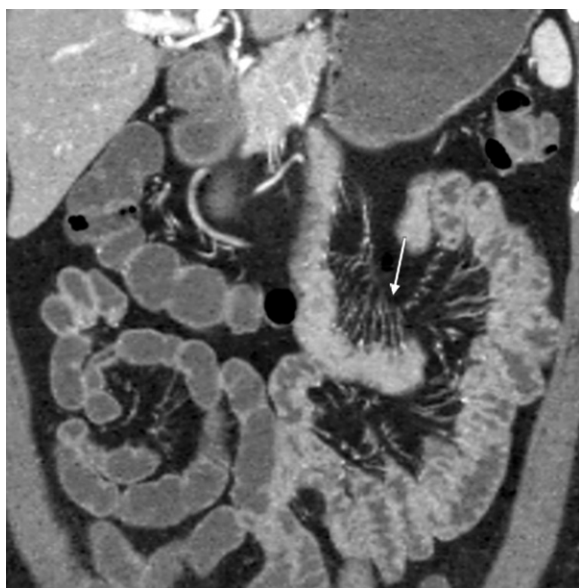


Figure 12. Celiac disease in a 41-year-old woman who presented with findings of anemia and episodes of epigastric pain. Coronal intravenous contrast-enhanced CT enterographic image shows engorgement of the mesenteric vessels (arrow), known as the comb sign.

by the edematous submucosa, and finally the hyperenhancing serosa (Fig 11) (31).

According to a study by Warshauer and Lee (32), intussusceptions can be recurrent, can be more than 3 cm in length, and can occur at different locations.

Intussusceptions in patients with celiac disease are thought to be related to intestinal atony (33) and, rarely, to polyps or masses (14). A transient self-limited intussusception most likely occurs when it is less than 3 cm long and in the absence of a detectable lead point mass or small bowel obstruction. However, because patients with celiac disease are at risk for devel-

oping gastrointestinal malignancies, each intussusception should be carefully examined for a lead point mass (3).

Mesenteric Vascular Engorgement.—Inflammation of the small bowel increases blood flow to the affected segment of bowel. This is seen at CT or MR enterography as mesenteric vascular engorgement (Fig 12). When multiple adjacent mesenteric vessels are engorged and distended, this gives the appearance of a comb (1). The finding of mesenteric hyperemia is not specific to celiac disease and can be seen in a number of enteric pathologic conditions.

Extraintestinal Findings

Mesenteric Adenopathy.—T- and B-cell lymphocyte stimulation leads to lymph node follicular hyperplasia (1). At imaging, this manifests as mesenteric lymphadenopathy (short axis >2 cm) (Fig 13). Mesenteric lymphadenopathy is the most frequent extraintestinal finding and is reported in up to 42% of patients (3). As a result of chronic lymphocyte stimulation, patients with celiac disease are more likely to have an increased risk of lymphoma (1).

One unique finding in patients with celiac disease is the presence of hypoattenuating or cavitating lymphadenopathy detected at CT enterography. Cavitated lymph nodes are found in patients with advanced disease, and their incidence has been reported in up to 12% of patients (34).

Hyposplenism.—Celiac disease is the most common gastrointestinal entity associated with hyposplenism (35). In chronic or severe cases of celiac disease, a decreased spleen size or splenic atrophy may be seen at CT and MR enterography. Hyposplenism may be present in 30%–50% of adults with celiac disease and mainly affects patients with chronic gluten exposure (Fig 14) (36). It is usually irreversible and does not normalize after cessation of gluten intake. It is caused by an

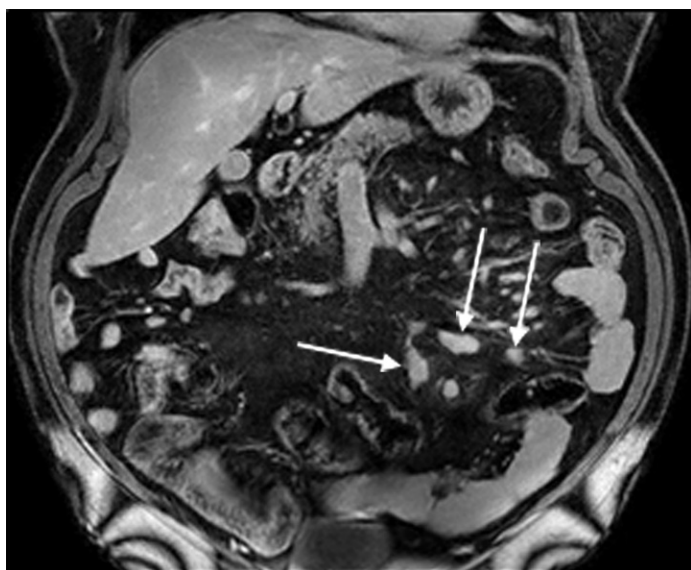


Figure 13. Celiac disease in a 66-year-old man who presented with abdominal pain and bloating. Coronal intravenous contrast-enhanced fat-suppressed T1-weighted MR enterographic image shows multiple mesenteric lymph nodes that are enlarged (arrows).

impaired immune function with a reduced number of immunoglobulin-M memory B cells, the defective activity of opsonizing molecules, and the depletion of regulatory T cells with a decreased number of marginal zone B cells, which lead to a reduction in spleen size (37,38).

Splenic atrophy is often associated with the presence of cavitated lymph nodes and, when seen together, can increase diagnostic specificity for celiac disease (14). Splenic atrophy may complicate celiac disease and other intestinal disorders such as Crohn disease, ulcerative colitis, and tropical sprue (33). The severity of hyposplenism increases with age and the duration of gluten exposure, although it is not associated with an increased risk of malignancy (39).

Ascites.—Although ascites is not a common finding in celiac disease, it may signify severe malabsorption and hypoproteinemia (33). It is important to look for other related findings of hypoalbuminemia such as anasarca, pleural effusions, subcutaneous edema, and low attenuation of the liver due to edema (Fig 9C, 9D) (33).

Complications

Ulcerative Jejunoileitis

Patients with refractory celiac disease may develop chronic ulcerations affecting the jejunum and ileum, usually more severe in the jejunum. The colon and duodenum are rarely affected. Clinically, these ulcers can produce abdominal pain and distention, weight loss, fever, and malabsorptive diarrhea with protein loss. Imaging features include jejunal wall thickening, mural stratification, and multiple luminal-sided depressions representing mucosal ulcers (3). Edema and mural hyperenhancement due to inflammation may also be seen (14). Chronic penetrating ulcers and inflammation can lead to fibrosis and stricture (Figs 15–17) (3). Ulcer-

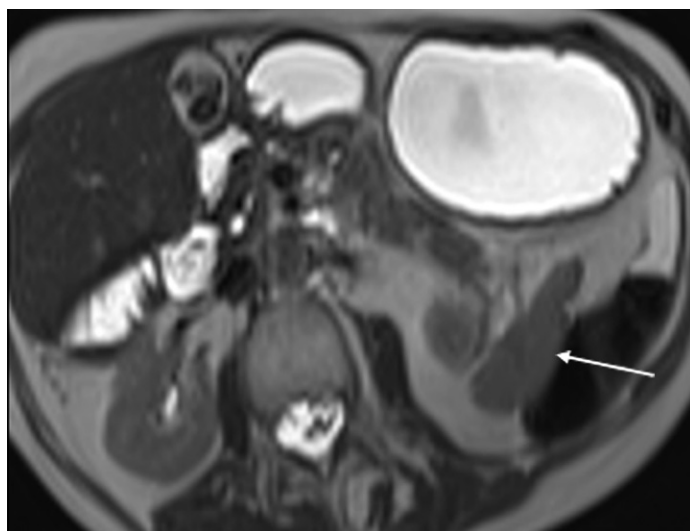


Figure 14. Refractory celiac disease in a 61-year-old woman. Axial T2-weighted MR enterographic image shows hyposplenism (arrow).

ations may be isolated, clustered, or more diffusely distributed throughout the jejunum (14). When present, ulcerative jejunoileitis should be further evaluated with endoscopy, as ulcerative jejunoileitis is considered a preneoplastic state, predisposing to the development of enteropathy-type T-cell lymphoma. Ulcerative jejunoileitis may lead to severe stricture or hemorrhage, perforation, and obstruction (3). Symptoms and radiologic features may develop acutely or over a more prolonged time (31).

Malignancy

Among the rarer complications of inadequately treated celiac disease is the potential of developing small bowel malignancies, most commonly lymphoma and adenocarcinoma (40). Therefore, clinical and imaging surveillance in symptomatic patients with celiac disease may lead to early diagnosis.

Although the exact cause and pathophysiology of celiac disease–associated malignancy is not clear, several proposed theories include increased mucosal lymphocytes, increased permeability to oncogenic factors, malabsorption of protective substances such as vitamins A and E, and an altered immune system (41).

Enteropathy-associated T-cell lymphoma (EATL) is the most common malignancy associated with celiac disease. Symptoms are often nonspecific, and imaging and clinical features can overlap with those of ulcerative jejunoileitis. Although EATL is the most common lymphoma to affect patients with celiac disease, diagnosis often occurs late in the disease course, and patients with EATL often have a poor prognosis. Patients may present with recurrent or persistent anemia, weight loss, abdominal pain, or obstruction, despite implementation of a gluten-free diet. EATL often involves a single longer segment of bowel and has a predilection for the ileum. Involvement of the bowel may be focal or diffuse.

Often, EATL may be difficult to detect with conventional imaging methods, and a high index of suspicion should lead to small bowel endoscopy or PET/CT to help make the

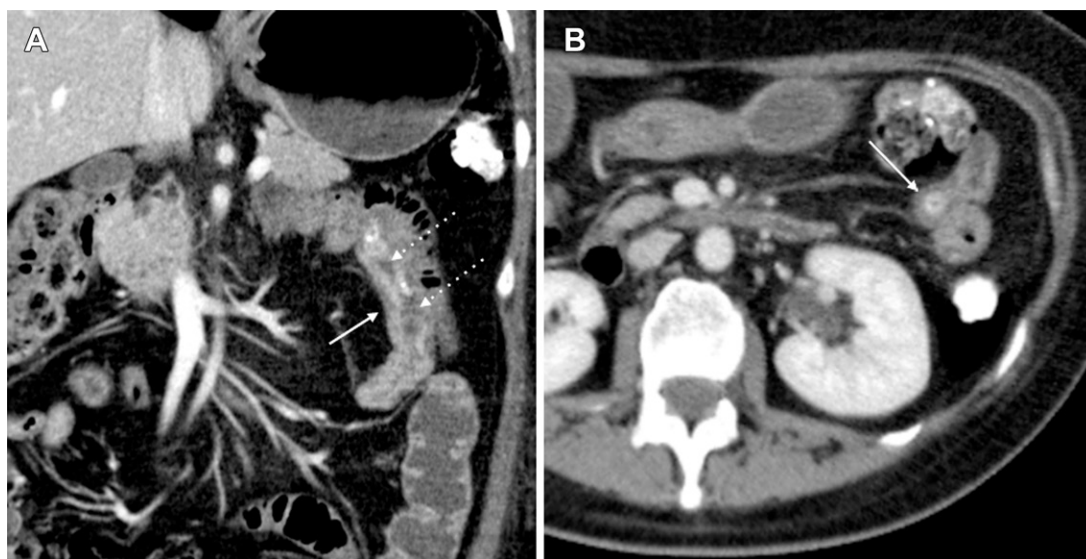


Figure 15. Ulcerative jejunoileitis in a 46-year-old woman with long-standing colicky abdominal pain. Coronal (A) and axial (B) intravenous contrast-enhanced CT images show thickening of the jejunal wall (solid arrow) and bilaminar stratified hyperenhancement due to inflammation and ulcerations (dashed arrows in A).

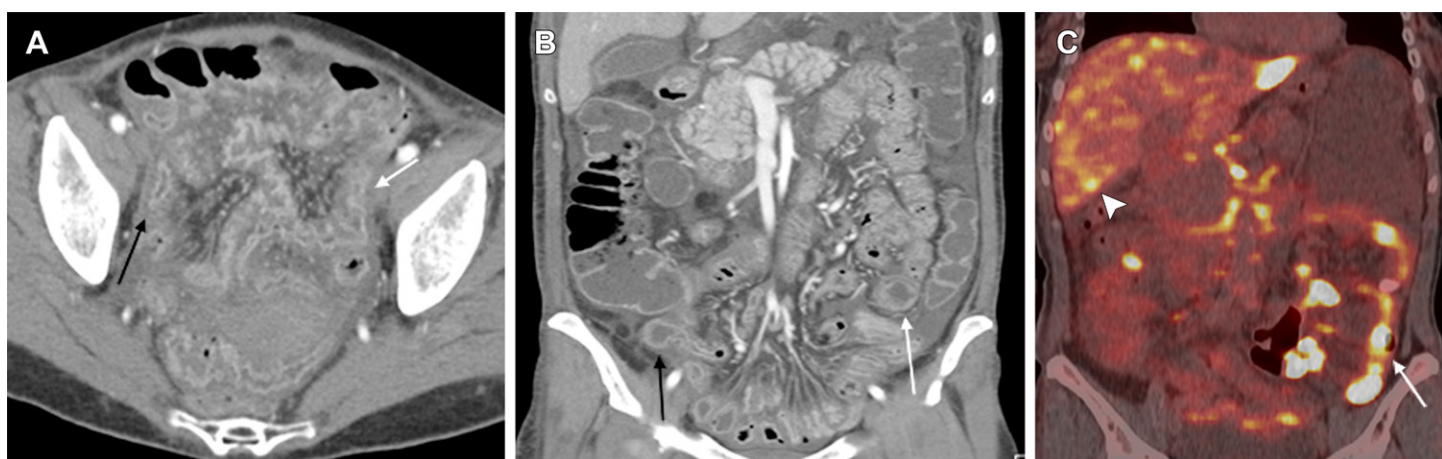


Figure 16. T-cell lymphoma associated with enteropathy in a 56-year-old woman with a known history of refractory celiac disease who presented for evaluation of abdominal pain. Axial (A) and coronal (B) contrast-enhanced CT enterographic images show diffuse intestinal edema, mural stratification, and thickening of the intestinal wall involving multiple loops of the jejunum and ileum (arrows), compatible with enteritis. In a patient with refractory celiac disease, ulcerative jejunoileitis may be considered. Biopsies demonstrated villous blunting and an abnormal T-cell infiltration. The patient then presented with weight loss and night sweats and underwent PET/CT. (C) Coronal fluorine-18 (^{18}F) fluorodeoxyglucose (FDG) PET/CT fusion image obtained 12 days later shows FDG avidity in the liver (arrowhead) and nodular areas of FDG avidity in the jejunum and ileum (arrows). Open intestinal resection demonstrated T-cell lymphoma associated with enteropathy.

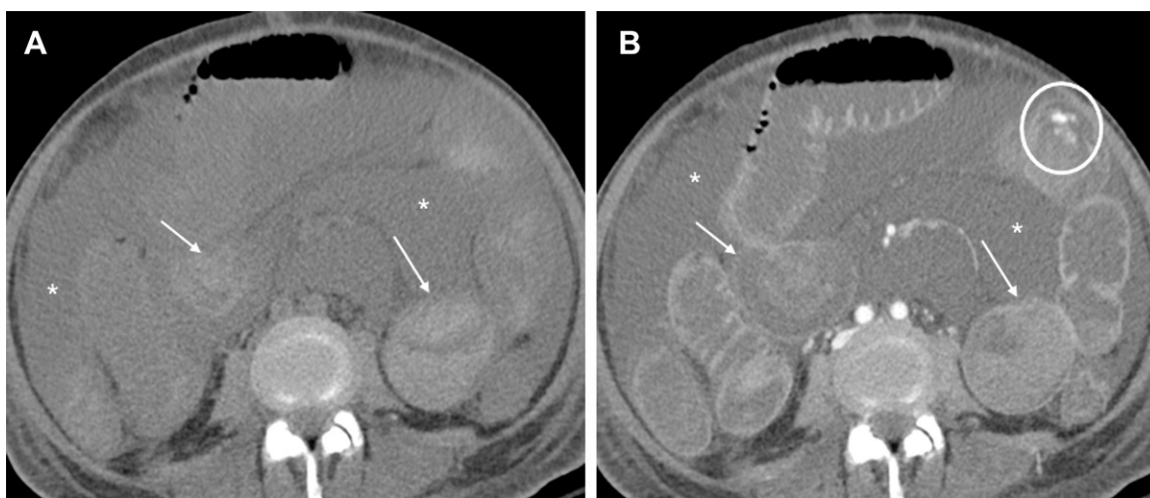
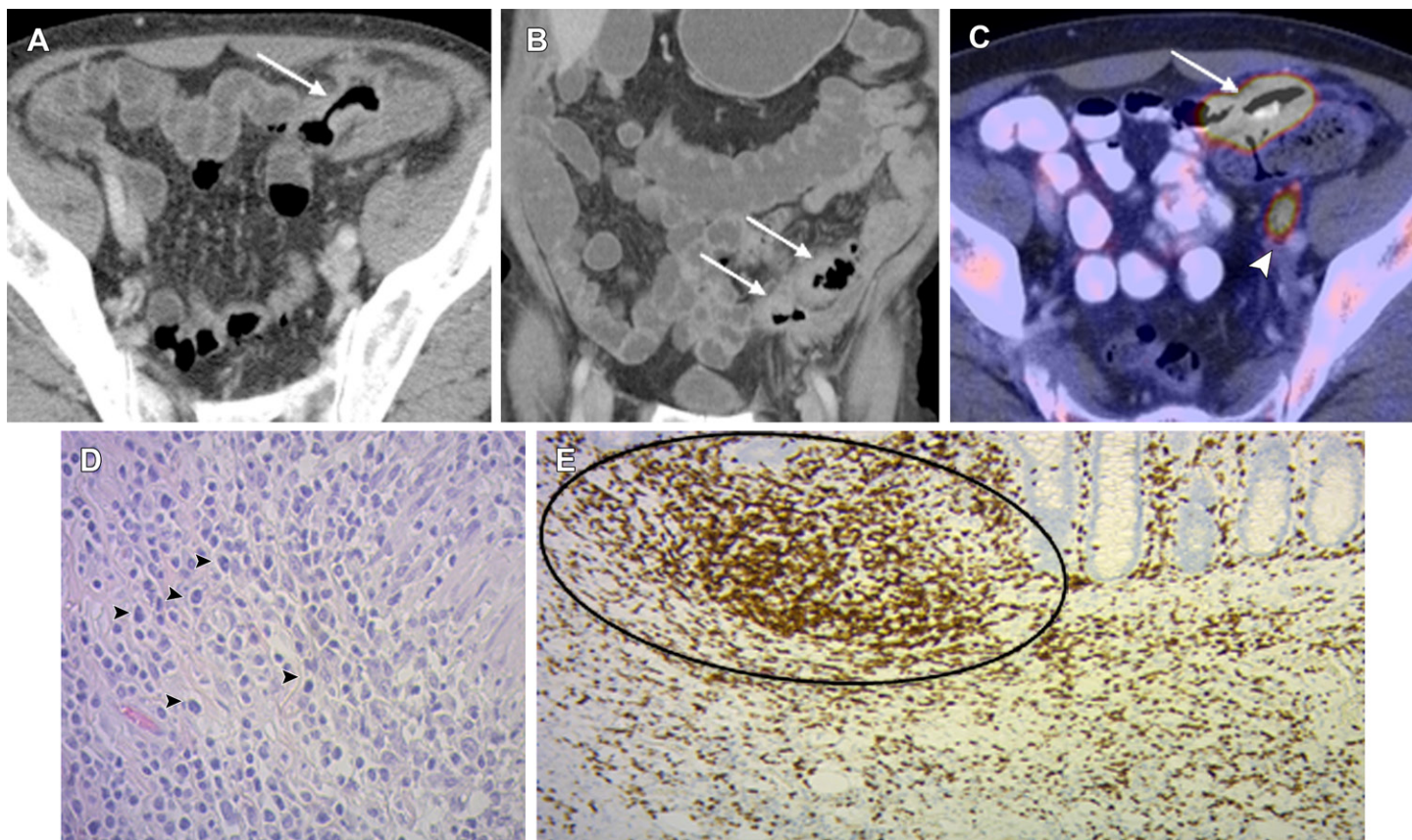


Figure 17. Ulcerative jejunoileitis and enteropathy-type T-cell lymphoma in a 56-year-old woman with celiac disease who presented with abdominal pain and hypotension (same patient as in Fig 16). Axial noncontrast (A) and arterial phase (B) CT images show active arterial extravasation from the ulcerative jejunoileitis (oval in B) and dilated blood-filled small bowel loops (arrows) and ascites (*).

Figure 18. Type II refractory celiac disease in a 68-year-old man with abdominal pain. **(A, B)** Axial **(A)** and coronal **(B)** intravenous contrast-enhanced CT enterographic images show asymmetric and segmental thickening of the jejunal wall, with soft tissues extending into the perienteric tissues (arrows). **(C)** Axial ^{18}F -FDG PET/CT fusion image shows increased uptake in the primary tumor (arrow) and a left internal iliac adenopathy (arrowhead). Jejunal resection was performed, and the results demonstrated T-cell lymphoma. **(D)** Photomicrograph of the histologic findings shows transmural involvement by angiocentrically distributed inflammatory cell infiltrate, which consists of lymphocytes with irregular nuclei and distinct nucleoli (arrowheads). (H-E stain; original magnification, $\times 4$.) **(E)** Photomicrograph using a CD3 staining technique shows a tumor composed of T cells (oval), compatible with T-cell lymphoma. (H-E stain; original magnification, $\times 40$.)



diagnosis (Fig 18). When seen, the imaging appearance of EATL is similar to that of other small bowel lymphomas and may appear as wall thickening and aneurysmal dilatation of the bowel. Although EATL may affect a long segment of bowel, obstruction is rarely seen. EATL is often associated with mesenteric adenopathy, and the involved lymph nodes may form a large bulky conglomerate or can demonstrate cavitation. One-third of patients with EATL have involvement of other organs, such as the liver or spleen, at initial diagnosis (14). Aneurysmal intestinal dilatation is an uncommon lymphoma-related finding that may cause bowel obstruction because of tumor replacement of the muscularis propria and infiltration of adjacent loops (42).

Patients with celiac disease are also at increased risk for developing nonlymphomatous malignancies, the most common of which is small bowel adenocarcinoma (Figs 19, 20). The jejunum is the most commonly affected site. Small bowel adenocarcinoma manifests as an irregular annular or semi-annular ulcerative mass with intra- and extraluminal growth and may be associated with an area of stricture that is sometimes described as an “apple core” lesion (30). Although it may be difficult to differentiate from small bowel lymphoma, adenocarcinoma is more often associated with small bowel

obstruction (14). In addition to small bowel adenocarcinoma, several studies have reported an increased risk of esophageal adenocarcinoma in these patients (43).

Cavitary Mesenteric Lymph Node Syndrome

Cavitary mesenteric lymph node syndrome (CMLNS) is a rare complication with uncertain causes and incidence, with at least 50 cases reported in the literature (44). CMLNS carries an extremely poor prognosis, with some studies demonstrating a 50% mortality rate (45). Clinically, patients with this pathologic condition present with malabsorption symptoms, even after a prolonged gluten-free diet, including diarrhea, anemia, weight loss, and fatigue (46). Histologically, the nodes have a large central cavity with a lipid or hyaline-type material. At imaging, CMLNS manifests as numerous enlarged mesenteric lymph nodes that vary in size (2–8 cm in diameter) and are centrally low in attenuation, signifying their centrally lipid or hyaline-type composition. Some lymph nodes may demonstrate a fat-fluid level, which is highly specific for CMLNS in celiac disease (Fig 21) (46). Findings of hyposplenism are commonly coexistent with CMLNS (46).

The differential diagnosis of this type of adenopathy includes lymphoma and mycobacterial infections. A useful

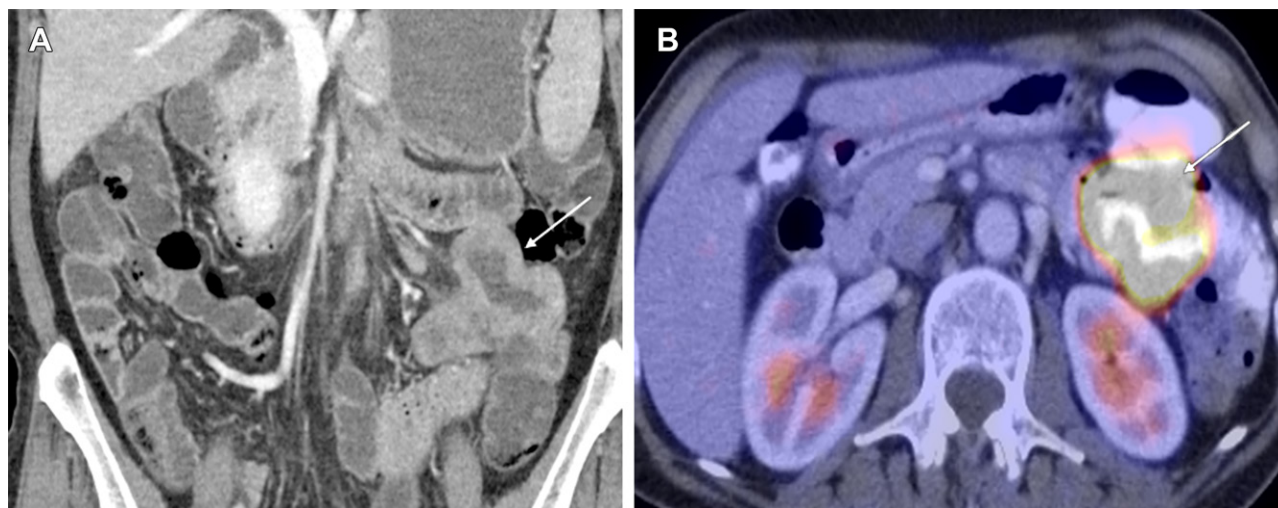


Figure 19. Jejunal adenocarcinoma in a 58-year-old woman with abdominal distention, anemia, and a positive fecal occult blood test result. **(A)** Coronal contrast-enhanced CT enterographic image shows asymmetric and focal thickening of the jejunal wall (arrow). **(B)** Axial PET/CT fusion image shows increased ¹⁸F-FDG uptake in the jejunal wall tumor (arrow).

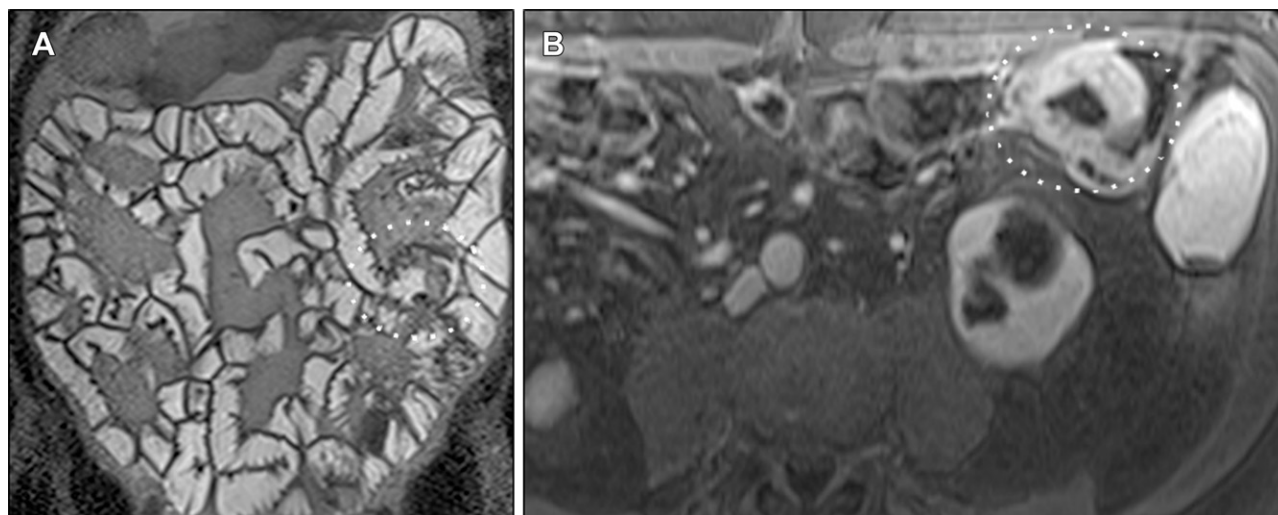


Figure 20. Adenocarcinoma in a 70-year-old man with celiac disease who presented for evaluation of refractory anemia. Coronal T2-weighted image **(A)** and axial intravenous contrast-enhanced fat-suppressed T1-weighted MR enterographic image **(B)** show increased diameter of a jejunal loop, aneurysmal dilatation, and associated bowel wall thickening (circle) representing signs of transmurular involvement by adenocarcinoma.

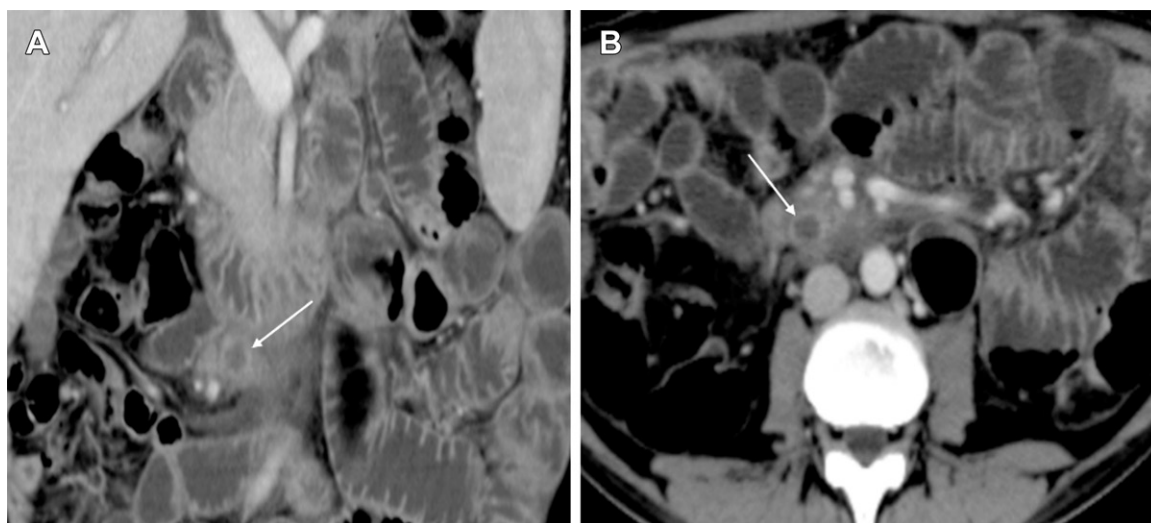


Figure 21. Cavitory mesenteric lymph node syndrome in a 40-year-old woman with untreated celiac disease who presented to the emergency department with epigastric pain and findings of anemia. Coronal **(A)** and axial **(B)** CT enterographic images with intravenous contrast agent administration show enlarged lymph nodes with a large central hypoattenuating cavitation component with lipid or hyaline material (arrow).

distinguishing feature is that in lymphoma, the lymphadenopathy is unlikely to be confined to the mesenteric nodes. In addition, low-attenuating lymphadenopathy in lymphoma is typically seen only in patients treated for lymphoma (47). Most patients with lymphoma present with splenomegaly, unlike patients with CMLNS, who have hyposplenism (48). Mycobacterial infections, especially tuberculosis, manifest with low-attenuating lymph nodes. However, clinical data and involvement of other organs are essential in differentiating these infections from CMLNS (48).

Treatment

Currently the only effective treatment available for celiac disease is a gluten-free diet, as it improves symptoms, can decrease antibodies, and has a partial protective effect against several complications (6). Once treatment is instituted, patients should be evaluated 4–6 weeks later by measuring levels of folate; vitamins A, D, E, and B12; copper; zinc; carotene; folic acid; ferritin; and iron and by performing a complete blood count, liver chemistry test, serologic test, and prothrombin time test (measured to detect possible vitamin K deficiency). Patients should then continue to be monitored for new or residual symptoms, assessment of compliance with the gluten-free diet, and detection of complications of celiac disease (49).

New therapies are focused on immunologic components at the level of gluten-reactive CD4 T cells rather than on inflammatory events. Teriflunomide, which proliferates gluten-reactive CD4 T cells, is of special interest. Other drugs under investigation are nanoparticles loaded with a gluten extract that lead to an inhibition of TG2 and decreased mucosal injury (50). The lifelong absence of wheat, rye, and barley has a significant psychologic and social impact on the patient, and dietary transgressions and poor adherence to the diet can occur (12). Therefore, it is important for these patients to have a multidisciplinary team that includes nutritionists, gastroenterologists, and psychologic support that can be of great help in the acceptance and management of celiac disease (6).

Conclusion

Patients with celiac disease present with nonspecific symptoms and a broad spectrum of imaging findings. However, the radiologist may help in the diagnosis of celiac disease by carefully examining the small bowel and extraintestinal findings. The most frequent findings are an inversion in the pattern of the small bowel folds and mesenteric lymphadenopathy. Small bowel imaging is also important in patients with a poor response or symptomatic recurrence after adherence to a gluten-free diet to detect complications and associated malignancies. The presence of ulcerative jejunoileitis, malignancy, cavitated mesenteric lymph nodes, or hyposplenism is an indicator of advanced disease.

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